$\boxed{\text{Apl}}$ Fie $A \in \mathcal{M}_n(\mathbb{C})$, $n \geq 2$, $\det A \neq 0$.

Arătați că: $(A^*)^* = (\det A)^{n-2} A$, A^* - m. adjunctă a lui A

Rez: $A^{-1} = \frac{1}{\det A} A^* \Rightarrow (A^*)^{-1} = \frac{1}{\det A^*} (A^*)^*$

$\Rightarrow A^{**} = (\det A^*)\,(A^*)^{-1}$

$A^* = (\det A)\,A^{-1} \Rightarrow \det A^* = (\det A)^n \underbrace{\det A^{-1}}_{\frac{1}{\det A}} = (\det A)^{n-1}$

$(A^*)^{-1} = \frac{1}{\det A} A$

Deci: $\boxed{(A^*)^* = (\det A)^{n-2} A}$

$\boxed{\text{Apl.}}$ Fie $A = (a_{ij})_{i,j=\overline{1,n}} \in \mathcal{M}_n(\mathbb{C})$, $\overline{A} = (\overline{a_{ij}})_{i,j=\overline{1,n}}$

a) $\det(\overline{A}) = \overline{\det(A)}$

b) $\det(A \cdot \overline{A}) = |\det(A)|^2 \geq 0$

Rez: a) $\overline{A} = (\overline{a_{ij}})$, $i,j = \overline{1,n}$

$\det \overline{A} = \sum_{\sigma \in S_n} \varepsilon(\sigma)\, \overline{a_{1\sigma(1)}}\, \overline{a_{2\sigma(2)}} \cdots \overline{a_{n\sigma(n)}} = \overline{\sum_{\sigma \in S(n)} \varepsilon(\sigma)\, a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}}$

$= \overline{\det A}$

b) $\det(A\overline{A}) = (\det A)(\det \overline{A}) = (\det A)(\overline{\det A}) = |\det(A)|^2 \geq 0$.

$z \cdot \bar{z} = |z|^2$

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$\boxed{\text{Apl}}$ Fie $A=(a_{ij})_{i,j=\overline{1,n}} \in \mathcal{M}_n(\mathbb{C})$ a.î. $a_{ij}=\overline{a_{ji}}$, $(\forall)\, i,j=\overline{1,n}$.

Arătați că: $\det A \in \mathbb{R}$.

Rez: $a_{ij}=\overline{a_{ji}}$, $(\forall)\, i,j=\overline{1,n} \iff A=\overline{A^T}$

$\Rightarrow \det A = \det \overline{A^T}$

Dar: $\det \overline{A^T} = \overline{\det A^T} = \overline{\det A}$

$\Rightarrow \det A = \overline{\det A}$

$\Rightarrow \det A \in \mathbb{R}$ q.e.d.

Reguli de calcul în t-un determinant

$A \in \mathcal{M}_n(\mathbb{C}) \rightsquigarrow \det A \in \mathbb{C}$

$\underline{n=1}$ $\det(a_{11}) = a_{11}$

$\underline{n=2}$ $\det\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22}\end{pmatrix} = a_{11}a_{22}-a_{12}a_{21}$

$\underline{n=3}$ $\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33}\end{pmatrix} = a_{11}a_{22}a_{33}+a_{12}a_{23}a_{31}+a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}-a_{12}a_{21}a_{33}-a_{11}a_{23}a_{32}$

$\Rightarrow$ Regula lui Sarrus

$\Rightarrow$ Regula Δ

Dezvoltarea după o linie sau o coloană

Fie $A=(a_{ij})_{i,j=\overline{1,n}} \in \mathcal{M}_n(\mathbb{C})$

$$\det A \overset{\text{dezv.}}{\underset{L_i}{=}} \sum_{j=1}^{n} a_{ij} \underbrace{(-1)^{i+j} A_{ij}}_{\text{complementul algebric al elem } a_{ij}}$$

$A_{ij} \to$ determinantul matricei care se obține din A, prin eliminarea liniei $\underline{i}$ și coloanei j

$$\det A \overset{\text{dezv.}}{\underset{C_j}{=}} \sum_{i=1}^{n} a_{ij} \underbrace{(-1)^{i+j} A_{ij}}_{\text{compl. algebric al elem } a_{ij}}$$

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$\boxed{Ap!}$ a) Calculați $\det \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ 0 & & & a_{nn} \end{pmatrix} \overset{not}{=} \Delta_1$

b) Calculați $\det \begin{pmatrix} a_{11} & & & 0 \\ a_{21} & a_{22} & & \\ \vdots & & \ddots & \\ a_{n1} & \cdots & \cdots & a_{nn} \end{pmatrix} \overset{not}{=} \Delta_2$

Rez: a) $\Delta_1 \underset{L_n}{\overset{dezv.}{=}} a_{nn}(-1)^{2n} \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n-1} \\ & a_{22} & \cdots & a_{2n-1} \\ & & \ddots & \vdots \\ 0 & & & a_{n-1,n-1} \end{vmatrix} \underset{L_{n-1}}{\overset{dezv.}{=}} a_{nn}\, a_{n-1\,n-1} \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n-2} \\ & a_{22} & \cdots & a_{2n-2} \\ & & \ddots & \\ 0 & & & a_{n-2\,n-2} \end{vmatrix}$

$= \ldots = a_{11} a_{22} \ldots a_{nn} = \prod_{i=1}^{n} a_{ii}$

{ determinantul unei matrice superior triunghiulare }.

b) Analog, prin dezv. succesive după $L_1, \ldots, L_n$ se arată că

$\Delta_2 = \prod_{i=1}^{n} a_{ii}$ { determinantul unei matrice inferior triunghiulare }

$\boxed{Ap!}$ Să se calculeze următorii determinanți:

a) $\Delta_1 = \begin{vmatrix} 1 & 2 & -1 & 4 \\ 3 & 1 & 4 & -5 \\ 2 & 0 & 1 & -1 \\ 6 & -5 & 4 & -4 \end{vmatrix}$

T b) $\Delta_2 = \begin{vmatrix} -2 & 5 & 0 & -1 \\ 1 & 0 & 3 & 7 \\ 3 & -1 & 0 & 5 \\ 2 & 6 & -4 & 1 \end{vmatrix}$

T c) $\Delta_3 = \begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 5 & 6 & 0 \\ 4 & 5 & 6 & 0 & 0 \\ 5 & 6 & 0 & 0 & 0 \end{vmatrix}$

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Rez: (V1) $\Delta_1 \underset{L3}{\overset{dezv.}{=}} 2\cdot(-1)^{3+1}\begin{vmatrix} 2 & -1 & 4 \\ 1 & 4 & -5 \\ -5 & 4 & -4 \end{vmatrix} + \cancel{0\cdot(-1)^{3+2}\begin{vmatrix} 1 & -1 & 4 \\ 3 & 4 & -5 \\ 6 & 4 & -4 \end{vmatrix}}$

$+ 1\cdot(-1)^{3+3}\begin{vmatrix} 1 & 2 & 4 \\ 3 & 1 & -5 \\ 6 & -5 & -4 \end{vmatrix} + (-1)\cdot(-1)^{3+4}\begin{vmatrix} 1 & 2 & -1 \\ 3 & 1 & 4 \\ 6 & -5 & 4 \end{vmatrix}$

$= 2(-32-25+16+80-4+40) + (-4-60-60$

$-\cancel{24}+\cancel{24}-25) + (4+48+15+6-24+20) =$

$= 2\cdot 75 - 149 + 69 = \underline{\underline{70}}$

(V2) $\Delta_1 = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 3 & -5 & 7 & -17 \\ 2 & -4 & 3 & -9 \\ 6 & -17 & 10 & -28 \end{vmatrix} = 1\cdot(-1)^2\begin{vmatrix} -5 & 7 & -17 \\ -4 & 3 & -9 \\ -17 & 10 & -28 \end{vmatrix}$

$= \begin{vmatrix} 5 & 7 & 17 \\ 4 & 3 & 9 \\ 17 & 10 & 28 \end{vmatrix} = 420 + 1071 + 680 - 867 - 784 - 450$

$= 2.171 - 2.101 = \underline{\underline{70}}$

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$\boxed{\text{Apl.}}$ Calculați $\det A$

① (a) folosind dezvoltarea după prima linie

(b) folosind regula lui Laplace prin dezv. după primele 2 linii.

$$A = \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 4 \\ 2 & 5 & 1 & -1 \\ -1 & -2 & 2 & 4 \end{pmatrix}$$

Rez: $\boxed{\text{Regula lui LAPLACE}}$

Fie $A \in \mathcal{M}_n(K)$, $1 \le p \le n$, $p \in \mathbb{N}$

$C_{ij} = (-1)^{i+j} M_{ij} \to$ compl. alg. al elem. a_{ij}

$C = (-1)^s M_c$, $s = (i_1 + i_2 + \dots + i_p) + (j_1 + j_2 + \dots + j_p)$

$\boxed{\text{Th. LAPLACE}}$ Determinantul matricei A e egal cu suma produselor minorilor de ordin p (ce se pot construi cu elem. a p linii (col) fixate ale m. A) prin compl. lor algebrici.

$\boxed{\text{C.P.}}$ $p = 1$

$(\forall)\ i = \overline{1,n}$: $\det A = a_{i1} C_{i1} + a_{i2} C_{i2} + \dots + a_{in} C_{in}$

(reg. de dezv. a det. matricei A după linia i).

$$\det A = \sum M \cdot M' = \sum \det(A_{IJ}) \cdot (-1)^{i_1 + \dots + i_p + j_1 + \dots + j_p} \det(A_{\bar{I}\bar{J}})$$

M minor de ordin p în A obț. din liniile $\{i_1, \dots, i_p\}$ și din p coloane

$1 \le p \le n$; $1 \le i_1 < i_2 < \dots < i_p \le n$

Obs: Suma are C_n^p termeni.

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Regula lui LAPLACE

Rez: (b) $\Delta = \det A =$

$$= \begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 1 & 3 & 4 \\ 2 & 5 & 1 & -1 \\ -1 & -2 & 2 & 4 \end{vmatrix} = (-1)^{1+2+1+2} \underbrace{\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}}_{=0} \begin{vmatrix} 1 & -1 \\ 2 & 4 \end{vmatrix} +$$

$$+ (-1)^{1+2+1+3} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} \begin{vmatrix} 5 & -1 \\ -2 & 4 \end{vmatrix} + (-1)^{1+2+1+4} \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} \begin{vmatrix} 5 & 1 \\ -2 & 2 \end{vmatrix}$$

$$+ (-1)^{1+2+2+3} \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} \begin{vmatrix} 2 & -1 \\ -1 & 4 \end{vmatrix} + (-1)^{1+2+2+4} \begin{vmatrix} 1 & 3 \\ 1 & 4 \end{vmatrix} \begin{vmatrix} 2 & 1 \\ -1 & 2 \end{vmatrix}$$

$$+ (-1)^{1+2+3+4} \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} \begin{vmatrix} 2 & 5 \\ -1 & -2 \end{vmatrix} =$$

$$= +0 - 1 \cdot 18 + 1 \cdot 12 + 1 \cdot 7 - 1 \cdot 5 + (-1) \cdot 1 =$$

$$= -18 + 12 + 7 - 5 - 1 = 19 - 24 = \underline{\underline{-5}}$$

Deci: $\boxed{\Delta = -5}$

(5) $\boxed{T}$ Arătați prin inducție matematică următoarea relație

$$V(a_1, \dots, a_n) = \begin{vmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_n \\ a_1^2 & a_2^2 & \dots & a_n^2 \\ \dots & \dots & \dots & \dots \\ a_1^{n-1} & a_2^{n-1} & \dots & a_n^{n-1} \end{vmatrix} = \prod_{1 \le i < j \le n} (a_i - a_j),$$

$(\forall)\ a_i \in \mathbb{C}$ {determinantul VANDERMONDE}

$\boxed{T}$ $\boxed{Ap.1.}$ Calculați $\det A$

a) folosind dezvoltarea după coloana 4

b) folosind regula lui Laplace prin dezv. după liniile $i_1 = 1, i_2 = 4$

$\cancel{i_1 = 1, i_2 = 3}$

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$$A = \begin{pmatrix} 1 & 3 & 0 & -2 \\ 2 & 1 & -1 & 3 \\ 1 & 2 & 3 & -1 \\ -1 & 4 & 0 & 1 \end{pmatrix}$$

Rez: b) $\Delta = \det A = \overbrace{\begin{vmatrix} 1 & 3 \\ -1 & 4 \end{vmatrix}}^{j_1=1,\, j_2=2} \cdot (-1)^{1+4+1+2} \cdot \begin{vmatrix} -1 & 3 \\ 3 & -1 \end{vmatrix}$

$$+ \overbrace{\underbrace{\begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix}}_{=0}}^{j_1=1,\, j_2=3} \cdot \cancel{(-1)^{1+4+1+3}} \begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} + \overbrace{\begin{vmatrix} 1 & -2 \\ -1 & 1 \end{vmatrix}}^{j_1=1,\, j_2=4} \cdot (-1)^{1+4+1+4} \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix}$$

$$+ \overbrace{\underbrace{\begin{vmatrix} 3 & 0 \\ 4 & 0 \end{vmatrix}}_{=0}}^{j_1=2,\, j_2=3} \cdot \cancel{(-1)^{1+4+2+3}} \cdot \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} + \overbrace{\begin{vmatrix} 3 & -2 \\ 4 & 1 \end{vmatrix}}^{j_1=2,\, j_2=4} \cdot (-1)^{1+4+2+4} \begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix}$$

$$+ \overbrace{\underbrace{\begin{vmatrix} 0 & -2 \\ 0 & 1 \end{vmatrix}}_{=0}}^{j_1=3,\, j_2=4} \cdot \cancel{(-1)^{1+4+3+4}} \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} =$$

$$= 7 \cdot (-8) + (-1) \cdot 5 + 11 (-1) \cdot 7 = -56 - 5 - 77 = \underline{-138}$$

Ag1: Considerăm următoarele matrice date pe blocuri:

1) $A = \begin{pmatrix} M_m & N \\ O & P_p \end{pmatrix} \to \begin{cases} M_{m \times m} \\ P_{p \times p} \\ N_{m,p} \end{cases} \Rightarrow \det A = \det M \cdot \det P$

2) $A = \begin{pmatrix} M_m & O \\ N & P_p \end{pmatrix} \to \begin{cases} M_{m \times m} \\ P_{p \times p} \\ N_{p \times m} \end{cases} \Rightarrow \det A = \det M \cdot \det P$

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3) $A = \begin{pmatrix} N & M_m \\ P_p & O \end{pmatrix} \to \begin{cases} M_{m\times m} \\ P_{p\times p} \\ N_{m\times p} \end{cases} \Rightarrow \det A = (-1)^{mp} \det M \cdot \det P$

4) $A = \begin{pmatrix} O & M_m \\ P_p & N \end{pmatrix} \to \begin{cases} M_{m\times m} \\ P_{p\times p} \\ N_{p\times m} \end{cases} \Rightarrow \det A = (-1)^{mp} \det M \cdot \det P$

$\boxed{\text{Obs:}}$ 1) $\underset{\text{Transpunere}}{\Rightarrow}$ 2)

Rez: 1) $A = \begin{pmatrix} M_m & N_{m\times p} \\ O_{p\times m} & P_p \end{pmatrix}$

Dem. că: $\boxed{\det A = \det M \cdot \det P}$

Calculăm $\det A$ cu regula lui LAPLACE, dezvoltând după primele "m" linii $(i_1 = 1, i_2 = 2, \dots, i_m = m)$.

Obținem: $\det A = \sum \underbrace{B}_{\substack{B \text{ este un} \\ \text{minor de ord. "m"} \\ \text{din } A \text{ determinat} \\ \text{de liniile } \{i_1, \dots, i_m\} \\ \text{și coloanele } \{j_1 < \dots < j_m\} \subset \{1, \dots, m+p\}}} \cdot \underbrace{B'}_{\substack{\text{compl. algebric} \\ \text{al lui } B}} =$

$= \det M \cdot \underbrace{(-1)^{2(1+2+\dots m)}}_{+1} \cdot \det P + 0$

Arătăm că: pt. $\{j_1, \dots, j_m\} \neq \{1, \dots, m\} \Rightarrow \underline{B' = 0}$

Minorii din B' corresp. liniilor $\{m+1, \dots, m+p\}$ și la p coloane $\{1, \dots, m+p\} \setminus \{j_1, \dots, j_m\}$

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Deoarece $\{j_1, \dots, j_m\} \neq \{1, \dots, m\}$

"m" elem $\Rightarrow (\exists)\ \underline{\underline{k}} \in \{1, \dots, m\} \setminus \{j_1, \dots, j_m\}$

Așadar, coloana "k" din matricea A este folosită în minorul B'

și deci e o coloană formată doar din 0 $\Rightarrow$ $\underline{B' = 0}$

În consecință: $\boxed{\det A = \det M \cdot \det P}$

Analog 3) și 4)

Determinanți VANDERMONDE

Fie $a_i \in K$, $(\forall)\ i = \overline{1,n}$, $n \geq 2$

$$V(a_1, \dots, a_n) \overset{not}{=} \begin{vmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_n \\ a_1^2 & a_2^2 & \dots & a_n^2 \\ \vdots & & & \\ a_1^{n-1} & a_2^{n-1} & \dots & a_n^{n-1} \end{vmatrix} \rightarrow \text{det. VANDERMONDE}$$

- Dem. că: $V(a_1, \dots, a_n) = \prod_{1 \leq j < i \leq n} (a_i - a_j)$: $P(n)$

Dem: $\rightarrow$ Inducție după $\underline{\underline{n}}$

I $P(2)$ $V(a_1, a_2) = \begin{vmatrix} 1 & 1 \\ a_1 & a_2 \end{vmatrix} = a_2 - a_1 \Rightarrow P(2)$ adev.

II Pp. $P(n-1)$ adev. $\Rightarrow P(n)$ adev.

$$V(a_1, \dots, a_n) = \begin{vmatrix} 1 & 0 & \dots & 0 \\ a_1 & a_2 - a_1 & \dots & a_n - a_1 \\ a_1^2 & a_2^2 - a_1^2 & \dots & a_n^2 - a_1^2 \\ \vdots & \vdots & & \vdots \\ a_1^{n-1} & a_2^{n-1} - a_1^{n-1} & \dots & a_n^{n-1} - a_1^{n-1} \end{vmatrix}$$

$$\begin{cases} C_2' \rightarrow C_2 - C_1 \\ C_3' \rightarrow C_3 - C_1 \\ \vdots \\ C_n' \rightarrow C_n - C_1 \end{cases}$$

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Știm că: $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$

Obs: De pe fiecare coloană C_i scoatem factor comun $\boxed{a_i - a_1}$,

În continuare, dezv. după L_1 și obținem: pt. $(\forall)\ i = \overline{2,n}$

$$V(a_1, \dots, a_n) = (a_2 - a_1) \dots (a_n - a_1) \begin{vmatrix} 1 & \dots & 1 \\ a_2 + a_1 & \dots & a_n + a_1 \\ a_2^2 + a_1 a_2 + a_1^2 & \dots & a_n^2 + a_1 a_n + a_1^2 \\ \vdots & & \vdots \\ a_2^{n-2} + a_2^{n-3} a_1 + \dots + a_1^{n-2} & \dots & a_n^{n-2} + a_n^{n-3} a_1 + \dots + a_1^{n-2} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \dots & 1 \\ a_2 & \dots & a_n \\ \vdots & & \\ a_2^{n-3} & \dots & a_n^{n-3} \\ a_2^{n-2} & \dots & a_n^{n-2} \end{vmatrix} \cdot (a_2 - a_1) \dots (a_n - a_1) =$$

$L_2' = L_2 - a_1 L_1$

$\vdots$

$L_{n-2}' = L_{n-2} - a_1 L_{n-3}$

$L_{n-1}' = L_{n-1} - a_1 L_{n-2}$

$$= (a_2 - a_1) \dots (a_n - a_1) V(a_2, \dots, a_n) \underset{P(n-1)}{\overset{\text{Ip. de ind}}{=}}$$

$$= (a_2 - a_1) \dots (a_n - a_1) \prod_{2 \le j < i \le n} (a_i - a_j) = \prod_{1 \le j < i \le n} (a_i - a_j) \Rightarrow P(n) \text{ adev.}$$

Obs: Fie $a_i \in K$, $(\forall)\ i = \overline{1,n}$
K corp com.

$\Rightarrow$ 1) $V(a_1, \dots, a_n) = 0 \Leftrightarrow (\exists)\ 1 \le i \ne j \le n$ a.î. $a_i = a_j$

2) $V(a_1, \dots, a_n) \ne 0 \Leftrightarrow a_1, \dots, a_n$ distincte

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